

6. (a) Explain the methods used by the food industry to preserve food.

[6 QER]

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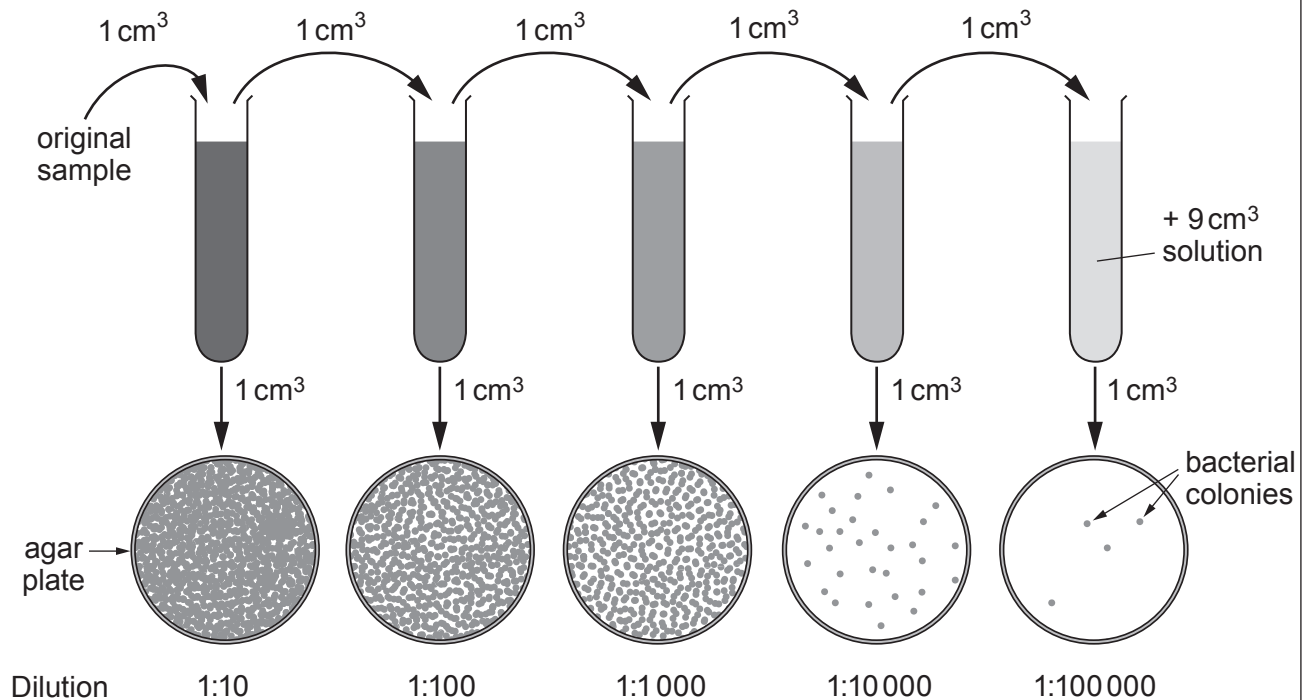
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- (b) Whenever an outbreak of food poisoning occurs, environmental health officers will carry out an investigation including testing samples of food for the presence of bacteria. To ensure a countable agar plate a series of dilutions is carried out as shown in the diagram below. The number of bacterial colonies on each plate is counted.



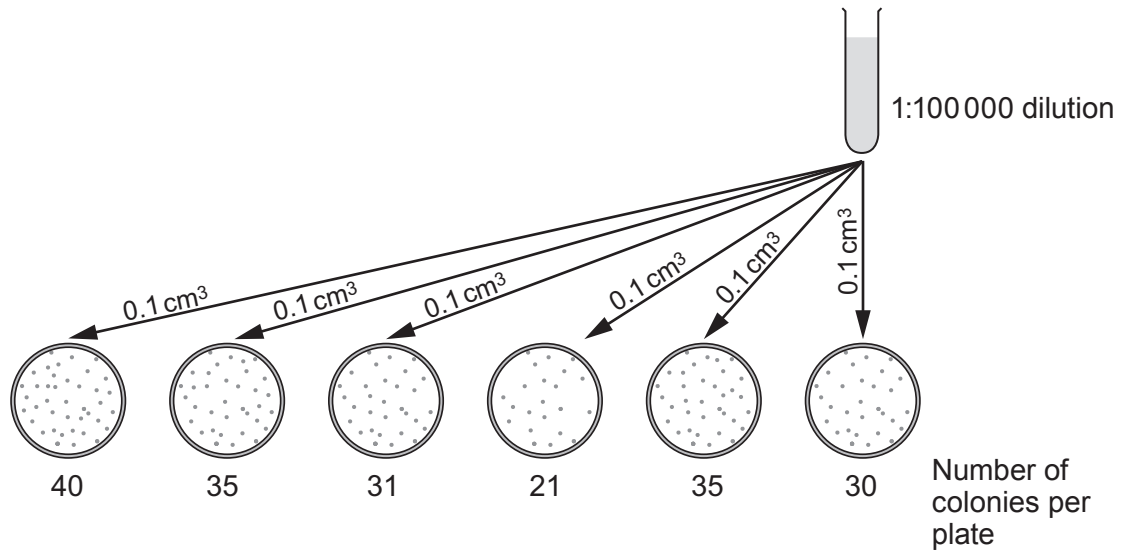
- (i) Suggest which dilution in the diagram above will provide the best set of results **and** give a reason for your answer. [2]

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- (ii) In another investigation, six agar plates were prepared by adding 0.1 cm^3 of solution from a 1:100 000 dilution. The agar plates were left for 72 hours. The results are shown below.



Use the information above to calculate the number of bacteria/ cm^3 in the original **undiluted** sample. [4]

Number of bacteria = / cm^3

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